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TOPICAL REVIEW

Toward Secure and Reliable IoT Systems: A Comprehensive Review of Formal Methods Applications

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ABSTRACT The exponential growth of Internet of Things (IoT) applications poses substantial obstacles in guaranteeing their dependability, safety, and effectiveness. Formal methods provide precise mathematical techniques for modeling, verifying, and analyzing these systems, effectively addressing important aspects of their creation and implementation. This review provides a thorough examination of the utilization of formal methods in the field of IoT, focusing on recent research published in well-regarded academic journals and conferences. We examine different formal methodologies, such as temporal logic, automata theory, theorem proving, and formal specification languages, emphasizing their efficacy in tackling difficulties peculiar to IoT systems. The review assesses the benefits and constraints of these methodologies, contrasts various approaches, and integrates essential discoveries. Key contributions include a comprehensive overview of formal methods applications, identification of critical challenges, evaluation of the effectiveness of different techniques, and proposals for future research directions. By highlighting emerging trends and future research paths, this review underscores the significance of formal approaches in enhancing the safety and resilience of IoT systems. This review seeks to serve as a beneficial reference for researchers and practitioners looking to improve IoT applications through formal verification and analysis techniques.

INDEX TERMS Formal methods, Internet of Things (IoT), reliability in IoT, model checking, theorem proving, formal specification, hybrid approaches, verification and validation, IoT system analysis, IoT safety and privacy.

I. INTRODUCTION

Applications for the Internet of Things (IoT) have proliferated quickly, transforming home automation, industrial automation, smart cities, and healthcare among other areas [1]. IoT devices in home automation, such as security cameras, lighting systems, and thermostats, provide consumers more protection, energy efficiency, and convenience [2]. Significant cost savings and productivity gains result from industrial automation's use of IoT to optimize production lines, streamline manufacturing processes, and track

equipment health [3]. Through linked systems like smart grids and intelligent transportation systems, smart cities use Internet of Things technologies to effectively manage resources, ease traffic congestion, and improve public safety [4], [5]. Remote patient monitoring, real-time health data collecting, and customized treatment plans made possible by Internet of Things applications in the healthcare industry greatly enhance patient outcomes and save healthcare costs [6], [7], [8]. Although the Internet of Things has vast potential and many applications, it faces significant security, efficiency, and reliability challenges [9]. In particular, these challenges are critical in applications like industrial automation and healthcare, where malfunctions, security

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breaches, or performance failures can have serious consequences [10], [11].

To address these concerns, formal methods provide rigorous mathematical solutions for the specification, development, and verification of systems and software [12]. Formal methods describe system behavior through mathematical models, allowing precise characterization and deep analysis [13]. These methods are invaluable for identifying contradictions, ambiguities, and inconsistencies early in the system specification phase, thus preventing errors and ensuring reliability. Formal approaches can help developers improve the security, dependability, and resilience of Internet of Things systems so that they satisfy strict functional and non-functional criteria [14], [15]. Formal methods include a range of techniques, such as formal specification languages, theorem proving, and model checking, each of which is designed to address specific aspects of system design and verification [16]. These approaches offer a structured framework for systematically verifying that IoT systems operate as intended under all circumstances, reducing the risk of malfunctions and security breaches [17], [18].

In [19], a formal analysis of the ACE specification for cache-coherent systems-on-chip, demonstrating the crucial role of formal methods in ensuring the reliability and performance of critical systems. They used model checking to validate coherence protocols and identified optimization opportunities, emphasizing how formal methods can detect subtle bugs that lead to major system failures. This research highlights the importance of formal verification for building reliable hardware systems. Moreover, they utilized model checking techniques to validate the coherence protocols and identified potential areas for optimization within the cache system. The study underscored the effectiveness of formal verification in detecting subtle bugs that could lead to significant system failures, highlighting the necessity for rigorous formal approaches in hardware design and verification. Moreover, they investigated the use of a formal model to improve the verification of a cache-coherent system-on-chip. Their research demonstrated how formal methods could be effectively applied to verify complex hardware designs, ensuring their correctness and performance. By employing model checking and other formal verification techniques, they identified and addressed potential issues in the system, showcasing the critical role of formal methods in the development of reliable hardware systems. This study contributed to the growing body of evidence supporting the use of formal methods in hardware verification.

Similarly, the formal modeling and validation of micro smart grids based on the ReDy architecture help to create accurate models of smart grid components and validate their interactions [20]. Through formal verification techniques, the study ensured that the smart grid system met rigorous performance and reliability standards, demonstrating formal methods' effectiveness in complex energy systems. The study provided valuable insights into the application of formal methods in the design and verification of complex energy

systems, demonstrating their potential to enhance the dependability of smart grid operations.

Examining formal methods' application in IoT system modeling, verification, and analysis is the goal of this paper [21], [22]. An extensive summary of current studies published in esteemed academic journals and conferences is given in this study, which also shows how formal methods are used to address problems unique to the Internet of things [23], [24]. Analyzing automata theory, formal specification languages, and temporal logic among other formal approaches, this research evaluates how well they improve the security, performance, and dependability of Internet of Things systems. For example, automata theory is applied to simulate the dynamic behavior of Internet of Things systems and validate their state transitions, while formal specification languages provide exact syntax and semantics for articulating system needs. Real-time Internet of Things applications especially need the ability to specify time-dependent actions, which temporal logic makes possible [25]. Real-time IoT applications especially benefit from temporal logic, which enables precise specification of time-dependent actions. By means of this review, we hope to offer insightful analysis of the state of formal methods research in the Internet of Things, point up new directions for future study to enhance the development and implementation of reliable IoT systems.

In comparison to existing surveys on secure and reliable IoT systems, this review offers several distinct contributions. While prior works primarily focus on general approaches to IoT security and reliability, our survey specifically emphasizes the application of formal methods as a rigorous and mathematical approach to tackling the challenges of IoT system dependability. Additionally, we provide a detailed comparative analysis of various formal techniques—such as temporal logic, automata theory, theorem proving, and formal specification languages—evaluating their specific advantages and limitations in the context of IoT. Unlike other surveys that may cover formal methods only briefly, our review delves deeply into recent research and emerging trends in this area, offering an up-to-date assessment of advancements that are not covered in earlier works. Furthermore, we identify critical challenges that remain in the field, such as scalability, real-time performance constraints, and the integration of formal methods with existing IoT technologies. Finally, we propose future research directions that address these gaps, positioning our review as a valuable resource for both researchers and practitioners seeking to advance the application of formal methods in secure and reliable IoT systems. The main contributions of this paper are summarized as follows:

- A comprehensive review of formal methods applied in IoT systems, demonstrating how these methods enhance security, reliability, and performance through rigorous mathematical verification.
- A detailed comparative analysis of formal techniques, including automata theory, temporal logic, theorem

proving, and formal specification languages, showing how each addresses specific challenges in IoT systems.

- A focus on emerging IoT challenges, such as scalability, real-time performance constraints, and the integration of formal methods with current IoT technologies, providing insights on adapting these methods to meet critical issues.
- New directions for future research, proposing ways to advance formal methods in secure and reliable IoT systems while identifying gaps in the current literature.

The organization of this paper is as follows: Section II provides an overview of the Internet of Things (IoT), including the main challenges faced by IoT systems and a review of formal methods. It also describes the methodology used in selecting and assessing relevant literature. Section III presents the findings from the literature review, focusing on formal techniques such as formal specification and verification, model checking, theorem provers, and hybrid approaches in IoT system architecture. Section IV highlights the knowledge gaps in the current body of research, covering issues such as scalability, real-time performance, the integration of varied formal methods, adaptability, security and privacy, user-friendly tools, and emerging IoT technologies. Section V addresses the key challenges and offers future considerations for advancing formal methods in IoT. Finally, Section VI concludes the paper by summarizing the main contributions and outlining the future directions for research in formal approaches to IoT.

II. BACKGROUND AND METHODOLOGY

A. INTERNET OF THINGS (IoT) OVERVIEW

IoT devices span a wide range, from basic sensors to multifunctional systems, enabling applications such as smart thermostats, industrial automation, and remote healthcare monitoring, enhancing efficiency and decision-making in these fields [26], [27], [28]. The rapid advancement of wireless technology, reduced sensor costs, and cloud computing has accelerated the growth of IoT, providing unprecedented opportunities for innovation. Advances in wireless transmission, sensor cost reduction, and cloud computing resources have accelerated IoT device proliferation [29]. However, this expansion introduces challenges related to scalability, interoperability, and security, especially as IoT systems must manage large data volumes and complex device interactions without compromising performance. IoT ecosystems face interoperability issues due to the diversity of devices and protocols, and require standardized communication frameworks to ensure seamless integration [30].

In [31], the authors provided an overview of IoT architectures, enabling technologies, and challenges, emphasizing the need for robust formal verification techniques to address the complexities of IoT systems. The findings demonstrated the significant potential of formal methods to enhance IoT resilience and reliability, establishing a strong argument for their integration into IoT system design.

IoT systems must prioritize security and privacy due to sensitive data transmission and storage [32]. This section briefly introduces IoT's challenges and paves the way for discussing formal methods as a solution for ensuring the secure and reliable operation of IoT systems.

B. CHALLENGES IN IoT SYSTEMS

Scalability remains one of the primary concerns as the proliferation of IoT devices leads to large volumes of data, demanding systems capable of handling exponential growth without sacrificing performance [33]. Since IoT devices are growing so quickly, scalability is a major problem. Interoperability is another critical issue, given the diversity of devices and communication protocols that must work together seamlessly [34]. This calls for scalable communication protocols that can manage large data volumes and device interactions, as well as reliable infrastructure and effective data processing algorithms. Still another important issue is interoperability [35].

Security emerges as the most pressing challenge, given the sensitive nature of data transferred across IoT systems and their vulnerability to cyberattacks. Strong encryption, authentication, and access control systems are essential to safeguard privacy and data integrity. Reliability and resilience are equally crucial, especially in mission-critical applications such as healthcare and industrial automation, where failures can have serious consequences [36]. Figure 1 illustrates the importance ratings of various challenges in IoT systems. The focus on security and privacy, reflected in their high importance ratings, further underscores the need for formal methods that can rigorously validate IoT system behavior to ensure reliability and protect sensitive data.

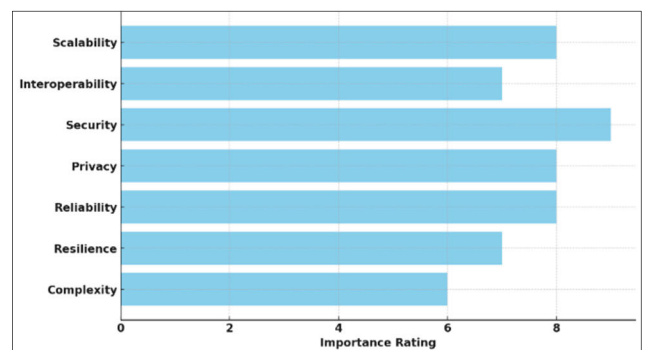


FIGURE 1. A bar plot showing the importance ratings of various challenges in IoT systems.

The delicate data being transferred and kept in Internet of Things systems makes security and privacy issues critical. Due to the presence of IoT devices in unsecured environments, they are particularly vulnerable to cyberattacks, which may result in unauthorized access, data breaches, or device manipulation. Strong security means putting encryption, authentication, and access control systems in place to safeguard privacy and data integrity [37]. Privacy concerns also

result from the ongoing gathering and processing of personal data, which calls for rigorous compliance with data protection laws and the creation of privacy-preserving technology. The resource allocation depicted in Figure 2 shows that the majority of resources are dedicated to security (25%) and privacy (20%), which aligns with their high importance ratings. Interoperability and reliability each receive 15% of resources, while scalability and resilience are allocated 10%, reflecting a balanced approach to address these challenges as IoT systems expand [38].

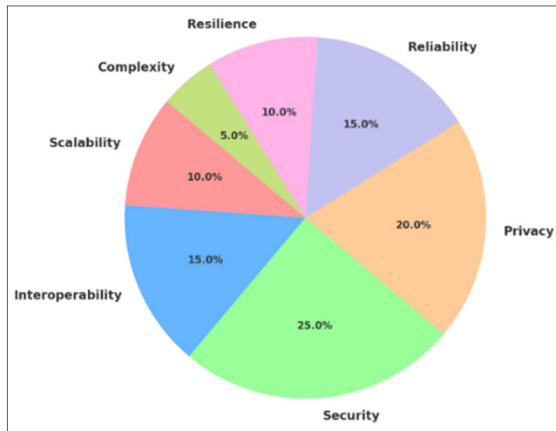


FIGURE 2. A pie chart illustrating the resource allocation percentages for addressing different challenges in IoT systems.

Particularly in vital applications like healthcare, industrial automation, and smart cities, reliability and resilience are critical to ensuring that IoT systems perform consistently across different environments and can withstand attacks or failures without compromising functionality [39]. Rugged testing, redundancy, and fault-tolerant design are all part of ensuring high dependability in order to reduce downtime and preserve continuous operation. IoT application safety and efficacy depend on resilience against physical faults and cyber-attacks. Figure 3 reveals that security is the most frequently addressed challenge in academic studies, with 50 studies focusing on this area. This reflects the critical nature of securing IoT systems and the extensive research efforts dedicated to this challenge. Privacy is also a major focus, with 40 studies highlighting the importance of safeguarding user data. Scalability, reliability, and resilience are addressed in 30, 35, and 25 studies respectively, highlighting the ongoing efforts to manage device growth, ensure dependable performance, and maintain system robustness. Interoperability and complexity receive comparatively less attention in the literature, with 20 and 15 studies respectively, suggesting a potential gap in research on device integration and system management.

In fact, IoT systems are by nature complex since they combine many technologies and are heterogeneous. The design, deployment, and maintenance of IoT systems require interdisciplinary expertise, involving hardware, software, networking, and data analytics. Handling this intricacy calls for

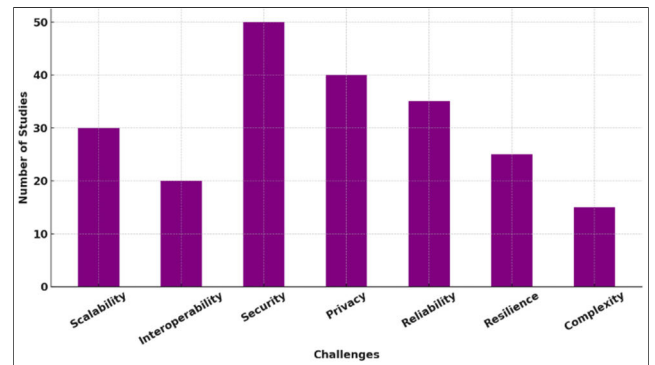


FIGURE 3. A histogram displaying the frequency of studies that address various challenges in IoT systems.

an all-encompassing strategy, from the beginning of design and development to deployment and continuing support. The intricacy of Internet of Things systems requires tools and approaches that make system design easier, deployment procedures faster, and monitoring and maintenance easier [40].

C. FORMAL METHODS OVERVIEW

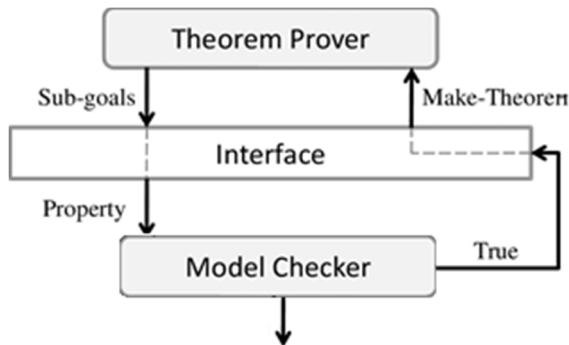
The specification, development, and verification of software and hardware systems are all areas that are addressed by formal methods, which require the application of mathematical concepts [16]. These techniques offer a rigorous framework for precisely defining complex systems, enabling in-depth analysis and verification to guarantee system correctness and reliability. There are many different approaches and tools that are included in formal methods, and each one is geared to a different aspect of system development [41].

Formal Specification Languages are used to describe the behavior and properties of systems in a precise and unambiguous manner. Syntax and semantics are provided by these languages, which include Z, VDM (Vienna Development Method), and B. These languages assist in the expression of system requirements and design [42]. Developers can use these languages to create detailed system models that undergo rigorous analysis to identify inconsistencies, ambiguities, and errors early in the development process, which helps prevent costly mistakes later [43]. Table 1 presents an overview of three widely used formal specification languages, highlighting their respective specification styles. Each language offers unique features and advantages for specifying and verifying system properties.

Specification Language Z is characterised by its sequential, property-oriented, and model-oriented specification styles. It provides a robust framework for clearly defining system behaviors and attributes. The B language is largely committed to the development of model-oriented specifications. It has a wide range of applications, including system modeling, which enables developers to design abstract representations of system components and the interactions between them. On the other hand, VDM is capable of supporting both process-oriented and model-oriented approaches

TABLE 1. Overview of widely used formal specification languages and their specification styles.

No	Specification Language	Specification Style
1.	Specification Language Z	Sequential Oriented, Property Oriented, and Model Oriented
2.	Specification Language B	Model Oriented
3.	Specification Language VDM	Process Oriented and Model Oriented

**FIGURE 4. Integration of theorem provers and model checkers.**

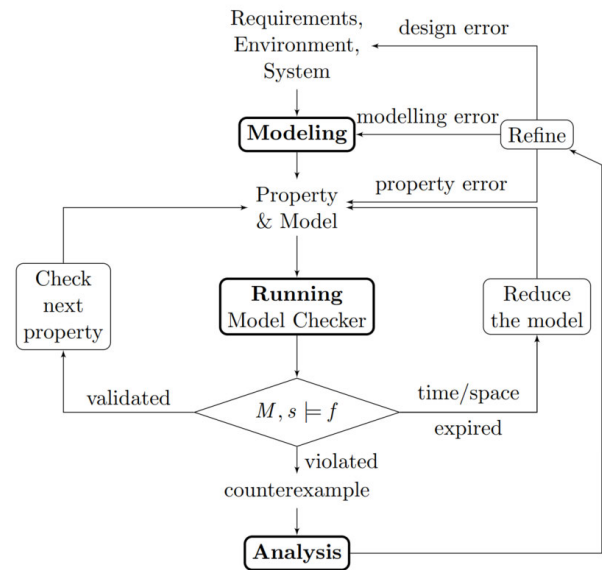
of specification. It provides a flexible method of formal specification, making it easier to describe both the processes of the system and the structural models of the system [44].

The process of proving a theorem entail employing logical reasoning in order to demonstrate that a system is right from the standpoint of its formal specifications. Coq and Isabelle are examples of theorem provers that provide assistance in the construction of formal proofs that establish whether or not a system complies to the properties that have been stated for it. Theorem proving is especially useful for verifying complex algorithms and protocols that cannot be easily validated through manual testing. A high level of assurance in the correctness of the systems is provided by theorem proving, which ensures that the systems meet the requirements that have been defined for them under any and all feasible circumstances [45]. Figure 4 illustrates the integration of theorem provers and model checkers through a common interface.

The integration of theorem provers and model checkers is a powerful approach to formal verification, combining the logical rigor of theorem proving with the automated efficiency of model checking. This combination allows for seamless interaction between the two tools, enhancing the verification process. Theorem provers provide detailed proofs of system correctness, while model checkers systematically explore system models to verify properties such as safety and liveness. This combined approach enhances the overall verification process, ensuring that IoT systems meet their specified requirements under all conditions [46].

One method that is used to determine whether a model of a system satisfies particular qualities, such as safety, liveness, and fairness, is called model checking. This method is an

automated methodology. SPIN, NuSMV, and UPPAAL are examples of model checkers that are designed to systematically investigate the state space of a system model in order to identify any property violations [47]. Figure 5 illustrates the process of model checking, which includes several steps to verify the properties of a system model. The process starts with modeling the environment and system, followed by running the model checker to validate properties.

**FIGURE 5. A flowchart representing the model checking process. The steps include modeling the environment and system, running the model checker, refining or reducing the model if necessary, and analyzing counterexamples if properties are violated.**

Model checking is particularly effective for detecting errors in concurrent and distributed systems, where the state space can be large and complex. Therefore, model checking is an extremely useful technique for assuring the dependability of a system because it has the capacity to detect minor defects that may be overlooked by more conventional testing approaches. The process begins with the modeling phase, where a detailed representation of the system and its environment is created. This model captures the essential properties and dynamics of the system, including its behavior and interactions. Accurate modeling is crucial, as any errors at this stage can propagate through the subsequent steps [48].

Once the model is constructed, the model checker is run. This tool systematically explores the state space of the system model, verifying whether the specified properties hold true. The input to the model checker includes the property to be verified and the system model itself. At the decision point, the model checker determines if the system model satisfies the specified properties [49]. This is denoted as $M, s \models f$. The possible outcomes are:

- Validated: The property is satisfied by the model.
- Violated: The property is not satisfied, and a counterexample is provided.

- **Time/Space Expired:** The model checker runs out of time or memory resources.

If a property is validated, the next property is checked. This step ensures comprehensive verification of all relevant properties of the system. If a modeling error is detected, the model is refined to improve accuracy. Similarly, if the model checker encounters time or space constraints, the model may be reduced by abstracting or simplifying certain aspects to facilitate efficient verification. Finally, the provided counterexample is analyzed to understand the nature of the violation. This analysis helps identify the root cause of the problem, guiding further refinement of the model [50].

D. METHODOLOGY

We conducted a systematic literature search to identify pertinent publications that discuss the utilization of formal techniques in modeling, validating, and analyzing Internet of Things (IoT) systems. The search was designed to ensure a comprehensive understanding of the application of formal methods in the rapidly evolving IoT landscape, focusing on high-quality and recent studies. We performed searches in several databases, including IEEE Access, IEEE Xplore, ACM Digital Library, Google Scholar, MDPI, ScienceDirect, and Scopus-indexed databases. As shown in Figure 6, the search for relevant papers was conducted across five major databases, with the number of papers retrieved from each database displayed below.

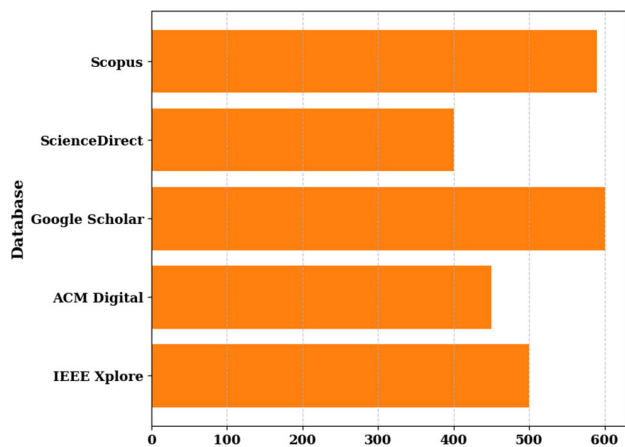


FIGURE 6. Overview of the search databases and the number of papers retrieved from each database during the systematic literature review.

The searches were conducted using the terms “formal methods,” “IoT,” “modeling,” “verification,” and “analysis.” Upon the initial search, a grand total of 2540 papers were found. In order to guarantee the excellence and pertinence of the chosen articles, many screening criteria were employed. The search terms were chosen to capture a broad range of formal techniques and their application to IoT systems, ensuring that no significant contributions were overlooked.

In order to guarantee the excellence and pertinence of the chosen articles, many screening criteria were employed.

The first level of screening involved reviewing titles and abstracts to remove papers that focused on unrelated topics, such as purely theoretical explorations of formal methods without IoT applications. The second level of screening applied more stringent criteria, including:

- **Relevance to IoT systems:** Papers had to demonstrate specific applications of formal methods in the context of IoT, focusing on modeling, verification, or analysis of these systems.
- **Publication quality:** Only peer-reviewed journal articles and conference papers were considered. Impact factor, citation count, and peer reviews were factored into the assessment of article quality.
- **Timeframe:** Only papers published after 2017 were considered, as this period marks the rise of newer formal methods tailored to the unique demands of IoT systems, such as increased scalability, real-time constraints, and enhanced security measures.

As a result, the list was trimmed to 1135 publications. Subsequently, papers were chosen for additional improvement based on two particular criteria: (1) the utilization of formal methods specifically designed for IoT systems; and (2) a focus on modeling, verification, or analysis of IoT applications rather than broad discussions on formal techniques. We also removed duplicates and papers that, upon further reading, did not meet the relevance criteria. After a thorough and demanding screening process, a final list of around forty eligible articles was obtained. This final set of studies was critically analyzed to ensure that they represent the most recent and relevant contributions to the intersection of formal methods and IoT. The literature selection process is depicted in Figure 7, highlighting how the number of papers was reduced at each screening stage.

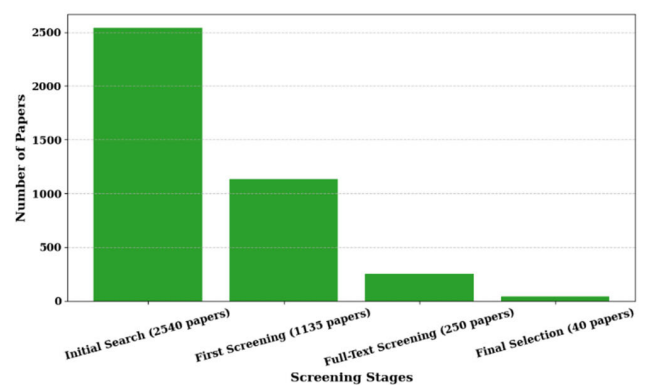


FIGURE 7. Literature selection process, showing the progression of papers from initial search to final selection through various screening stages.

The next stage of the review process involved a detailed examination of the selected papers. A methodical procedure for extracting and analyzing data was carried out, with a focus on understanding the different formal methodologies applied in IoT environments. The papers were classified based on the

formal methodologies employed and their particular applications in IoT environments. Three primary categories emerged from the review:

- Formal specification languages, which provide a precise way to define system requirements and behavior.
- Automata theory, used to model the dynamic behavior of IoT systems.
- Temporal logic, essential for specifying and reasoning about time-dependent properties in IoT applications.

Throughout the review process, we used reference management tools such as *Zotero* to organize and annotate articles, allowing for efficient extraction of key themes and trends. Challenges encountered during the review process included the need to filter out articles that discussed formal methods broadly without addressing IoT-specific challenges. In response, we refined our search terms and applied more focused inclusion criteria in subsequent rounds of review.

This rigorous methodology enabled us to provide a comprehensive assessment of the state of formal methods research in IoT, offering novel insights into the key trends and identifying areas for future study.

III. FINDINGS IN THE LITERATURE REVIEW

The systematic literature study on the utilization of formal methods in IoT systems provided valuable insights into the current level of research and practical implementation. This section consolidates the results obtained from the examined works, with a specific emphasis on the several classifications of formal methods, namely formal specification and verification, model checking, and theorem proving. A rigorous approach to describing, specifying, and testing these systems is provided by formal methods. This approach addresses a variety of issues that are inherent in the design and implementation of these systems. A literature survey on the application of formal methods in the IoT is presented in this part, with a focus on the most important studies, methodology, and developing trends. Several research have explored the utilization of formal approaches in various parts of IoT systems.

A. FORMAL SPECIFICATION AND VERIFICATION

Formal specification and verification play a crucial role in guaranteeing the dependability and accuracy of IoT systems. Multiple studies have showcased the utilization of these strategies in diverse circumstances. Presented below is an exhaustive analysis of significant research papers, arranged in chronological order from the earliest to the most recent.

A survey conducted in 2019 offered a thorough examination of formal verification methodologies for IoT systems, addressing the difficulties and possibilities in this domain [51]. The authors indicated crucial topics for future research to improve the implementation of formal approaches in IoT. The report highlights the crucial importance of

formal verification in progressing IoT technology. The study explores diverse formal approaches, including model checking, theorem proving, and formal specification languages. It investigates how these methods might be applied in various IoT scenarios and identifies the existing research and practice gaps. In the same year, a study devised a cloud-based framework for the formal verification of IoT tasks, with a specific emphasis on scheduling real-time remote tasks [52]. The tool emulates and assesses IoT operations in an intelligent environment, guaranteeing that time limits are adhered to in order to prevent dangerous circumstances, particularly in systems that are essential for safety and mission success. The suggested tool employs formal methods to verify the scheduling of jobs, guaranteeing compliance with stringent timing limitations. The study highlights the significance of validating temporal features.

In the upcoming year, a study introduced an innovative formal verification technique for state-based control modules in Cyber-Physical Systems (CPS). The process entails converting UML state machine diagrams into an abstract rule-based logical model [53]. This logical model is utilized for formal verification through model checking and for prototyping implementation on FPGA devices, guaranteeing coherence between the verifiable model and synthesizable VHDL code. This technique guarantees the proper and effective functioning of control modules, hence minimizing the likelihood of mistakes in CPS, which are essential components of Internet of Things (IoT) systems. Within the same framework, a review paper examines the current level of advancement in the formal verification of IoT protocols, emphasizing the difficulties and potential areas for future research. The authors conducted a comprehensive analysis of several formal verification methodologies and tools, highlighting the need of thorough verification in guaranteeing the security and dependability of IoT protocols [54]. The researchers analyzed various protocols frequently employed in IoT and assessed the merits and drawbacks of current verification techniques. They suggested potential enhancements and future research directions to bolster the security and efficiency of these protocols.

In 2021, the study's authors concentrated on the formal specification and verification of the MQTT protocol using PlusCal-2 [55]. The protocol underwent a transformation into a PlusCal-2 model, and formal verification was conducted using the TLA+ model checker to confirm its properties. Ensuring the accuracy of MQTT is crucial for facilitating communication in the Internet of Things (IoT). The researchers emphasized the appropriate application of formal approaches in communication protocols to prevent errors and ensure dependable data transmission in IoT networks.

In a separate paper, the authors introduced a formal model of a distribution management system for smart grids that utilizes the IoT [56]. The Nondeterministic Finite Automata (NFA) and Unified Modeling Language (UML) were

TABLE 2. Detailed Analysis of Key Studies on Formal Specification and Verification in IoT Systems.

Study	Year	IoT Application Field	Formal Technique Used	Primary Topic	Main Contribution	Limitations
[51]	2019	General IoT Applications	Formal Verification	Verification Methodologies	Examined verification methodologies for IoT systems.	Identified gaps in research and practice.
[52]	2019	Real-Time Tasks Scheduling	Formal Verification	Verification Tool Design	Designed a cloud-based framework for real-time task scheduling.	Complexity in scheduling real-time tasks.
[53]	2020	Control Modules in CPS	Formal Verification	State-Based Control Modules	Converted UML state machine diagrams into logical models for verification.	Complexity in converting UML diagrams.
[54]	2020	IoT Protocols	Formal Verification	IoT Protocols Verification	Analyzed and reviewed IoT protocol verification methodologies.	Scalability and adaptability issues.
[55]	2021	MQTT Protocol	Formal Specification and Verification	Protocol Verification	Transformed MQTT protocol into PlusCal-2 model for verification.	Resource-intensive transformation process.
[56]	2022	Smart Grids	Formal Modeling	Distribution Management	Constructed formal model for smart grids.	Complexity in modeling distribution systems.
[57]	2021	Safety-Critical IoT Systems	Formal Specification and Verification	Timing Behavior Verification	Presented formal specification and verification for timing behavior.	High computational requirements for timing analysis.
[58]	2022	IoT Architecture	Formal Verification	Correct-by-Construction Approach	Integrated formal verification at design stage for IoT architecture.	Resource-intensive verification process.
[59]	2023	IoT Operating Systems	Formal Specification and Verification	Scheduler Verification	Extracted, specified, and verified Contiki scheduler.	Complexity in verifying scheduler operations.
[60]	2023	Constrained Devices	Formal Verification	Multi-Factor Authentication	Verified multi-factor authentication scheme using temporal logic.	Scalability and adaptability issues.
[61]	2024	IoT Systems	Formal Verification	Proof System for IoT Systems	Presented proof system for CaT calculus to verify IoT systems.	Complexity in validating interactions.

utilized to construct the model, while the VDM-SL toolkit was employed for analysis. The study showcased the efficacy of formal approaches in guaranteeing the dependability and safety of energy distribution systems based on the Internet of Things (IoT). The text underscores the significance of precisely modeling and validating the functions of smart grids to avert faults and enhance performance. In [57], the authors presented a formal specification and verification approach for timing behavior in safety-critical IoT systems. The authors employed formal specification languages and model checking approaches to guarantee the fulfillment of temporal restrictions in IoT systems, a vital requirement for safety-critical applications. The research highlights the importance of doing accurate timing analysis and verification in systems where timing faults might result in severe and disastrous outcomes. During the same time frame, another work introduced a correct-by-construction method for formally verifying IoT architecture [58]. The technique incorporates formal procedures into the design process to guarantee that IoT systems are constructed correctly, hence greatly decreasing the probability of design defects. This approach integrates formal verification at an early stage in the design process of IoT systems. The study emphasizes the benefits of employing a correct-by-construction approach to develop dependable and secure IoT systems from the beginning.

A study conducted in 2023 introduces a method for extracting, formally specifying, verifying, and repairing the Contiki scheduler [59]. The Contiki scheduler is a lightweight operating system designed for IoT devices that operates based on

events. The researchers created a simplified representation of the scheduler's many modes of operation using a state machine. They then utilized Promela to formally describe and define this representation. The SPIN model checker was employed for formal verification, detecting and rectifying design problems in the scheduler. The study showcases the efficacy of formal approaches in enhancing the resilience of operating systems employed in IoT, guaranteeing their ability to handle events and tasks with utmost reliability. However, another study focused on the security issues in the IoT, specifically regarding the verification of limited-capacity devices [60]. The work employed temporal logic to formally verify that a multi-factor authentication scheme is free from deadlocks and satisfies recognized security standards. This emphasizes the significance of employing formal methods to validate security solutions beyond conventional encryption, guaranteeing strong authentication procedures in IoT environments.

Recently in [61], a paper presented a proof system for the CaT calculus, which is employed to formally verify IoT systems. The proof system facilitates the validation of intricate interactions in IoT contexts, guaranteeing the accuracy of system behaviors. The CaT calculus offers a strong foundation for validating interactions inside IoT systems, which is essential for preserving system integrity and operation. Each language offers unique features and advantages for specifying and verifying system properties. As detailed in Table 2, a comprehensive analysis of key studies on formal specification and verification in IoT systems reveals various

applications and contributions across different IoT fields. These studies encompass methodologies such as formal verification for general IoT applications and real-time task scheduling, state-based control modules in cyber-physical systems, and IoT protocol verification, highlighting their respective contributions and limitations.

The literature on formal specification and verification for IoT systems reveals a number of significant trends and insights. Formal methods, including model checking and theorem proving, have become increasingly popular as essential instruments for verifying system behavior and guaranteeing the security, reliability, and accuracy of IoT applications. These techniques are particularly effective in identifying potential security vulnerabilities and guaranteeing that IoT systems satisfy rigorous functional and non-functional requirements. Nevertheless, significant obstacles persist, particularly in the realm of scalability, as the complexity of large-scale IoT systems and their extensive state spaces continues to be challenging to manage. Furthermore, there is a requirement for formal verification tools that can accommodate the timing and performance constraints of real-time IoT systems, as the integration of formal methods into these systems is currently underexplored. Another substantial opportunity is the integration of automated tools and formal specification languages to optimize the verification process and minimize manual intervention.

By offering a comparative examination of the diverse formal methods employed in IoT systems, this paper contributes to the expanding corpus of knowledge in formal specification and verification. Our analysis provides a more comprehensive understanding of the techniques that are most appropriate for specific IoT scenarios by examining the strengths and limitations of various approaches, such as model checking, theorem proving, and formal specification languages. Additionally, our review identifies numerous gaps in the current literature, with a particular emphasis on the capacity to adopt formal methods to dynamic and heterogeneous IoT environments, as well as scalability and real-time performance. By addressing these gaps, we underscore the critical obstacles that must be surmounted in order for formal methods to be more widely adopted in IoT systems. Lastly, we suggest future research directions, with a particular emphasis on the necessity of scalable formal methods, real-time verification techniques, and the creation of user-friendly tools that are more easily assimilated by industry professionals.

B. MODEL CHECKING

Model checking is an essential formal verification technique employed to examine the state space of IoT systems in order to guarantee that they satisfy specified safety, security, and reliability criteria. This technique is especially beneficial in the IoT arena because of the intricate interactions among multiple devices and protocols. This section provides a comprehensive assessment of prominent research that have explored the utilization of model checking in the context of the Internet of Things (IoT). The evaluation focuses on

the methodology employed and the significant conclusions obtained from each study.

In one study [62], the researchers investigate the use of model checking to ensure the accuracy and efficiency of Trickle-based dissemination protocols in the Internet of Things (IoT). The methodology entails constructing formal models of the Trickle algorithm and employing model checking to verify its correct operation under different network conditions. The results illustrate the efficacy of the protocol and highlight possible areas for enhancement.

Similarly, resiliency and sustainability of in-vehicle network for real-time authenticity are assessed through model checking in a work published in 2021. This study utilizes model checking to guarantee the robustness and longevity of in-vehicle networks, specifically focusing on the CAN protocol. The methodology involves creating formal models of the CAN protocol using timed automata and use model checking to confirm its resilience against message flooding and bus-off assaults. The study establishes the necessary circumstances for the protocol's ability to withstand and maintain its effectiveness, which have been confirmed by rigorous analysis using model verification and experimentation [63].

In the context of IoT security, another study introduced a framework for conducting security analysis of IoT systems by use model checking to validate their safety and security attributes [64]. The process entails constructing a formal model of the IoT system and employing model checkers such as SPIN to scrutinize potential security risks and validate the system's compliance with specified security features. The efficacy of the framework is evidenced by multiple case studies, showcasing its capacity to identify and address security issues in IoT environments.

In this context, other researchers focused on investigating the behavior of sensors in IoT systems through the application of statistical model checking is the focus of this study. The primary objective of this study is to employ statistical model checking for the purpose of understanding and evaluating the performance of sensors within Internet of Things (IoT) systems [65]. The methodology combines learning algorithms with statistical model checking methods to assess the dependability and efficiency of sensor networks. The study utilizes sensor behavior modeling and extensive simulations to identify trends and anomalies, thereby insuring the system's resilience and dependability under a range of conditions.

The HSAS-MD Analyzer is a cutting-edge security analysis system that combines model-checking techniques and deep learning to detect malware in IoT applications. This research presents a novel approach to detecting malware in IoT applications by integrating model checking with deep learning in a hybrid security analysis system [66]. The process entails transforming the source code of an IoT application into a format compatible with a model checker. It then identifies any aberrant behaviors and verifies the application's behavior using model checking. The deep learning component improves detection accuracy by

TABLE 3. Review of methodologies and applications of model checking in IoT systems.

Study	Year	IoT Application Field	Formal Technique Used	Primary Topic	Main Contribution	Limitations
[62]	2020	Trickle-based Dissemination Protocols	Model Checking	Verification	Verified correct operation of Trickle algorithm.	Possible areas for enhancement.
[63]	2021	In-Vehicle Network	Model Checking	Resiliency and Sustainability	Guaranteed robustness and longevity of in-vehicle networks.	Resource-intensive formal models.
[64]	2021	IoT Security	Model Checking	Security	Validated safety and security attributes of IoT systems.	Scalability and adaptability issues.
[65]	2022	Sensor Networks	Statistical Model Checking	Performance Analysis	Assessed dependability and efficiency of sensor networks.	High computational requirements.
[66]	2022	Malware Detection in IoT Apps	Model Checking, Deep Learning	Security Analysis	Detected malware in IoT applications.	Complexity in integrating deep learning.
[67]	2023	IoT Systems Security	Model Checking, Self-Adaptation	Security and Adaptation	Enhanced security and flexibility of IoT systems.	Complexity in integrating multiple techniques.
[68]	2023	IoT Systems	Model Checking	Survey on Verification and Validation	Reviewed various formal verification and validation techniques.	Resource-intensive surveys.

categorizing possible threats according to acquired patterns. The system exhibits exceptional accuracy, precision, recall, and F-measure in identifying malicious activity.

In a conceptual proposal, researchers integrated data-driven security, model checking, and self-adaptation for IoT systems using BIP (Behavior, Interaction, Priority) components. The proposed model aims to combine data-driven security, model checking, and self-adaptation techniques. This approach combines the use of data-driven security, model checking, and self-adaptation to enhance the functionality of IoT systems [67]. The integration is achieved through the utilization of BIP (Behavior, Interaction, Priority) components. The process entails the creation of formal models for IoT components, doing model checking to validate security features, and implementing self-adaptation methods to dynamically fix discovered security vulnerabilities. The strategy seeks to improve the overall security and flexibility of IoT systems.

In a related study, a survey paper reviewed various formal verification and validation (FV&V) techniques applied to IoT systems. This paper provides an overview of different formal verification and validation (FV&V) methods used in the context of Internet of Things (IoT) systems [68]. The methodology comprises a comprehensive examination of existing literature, classification of various FV&V methods, and evaluation of their efficacy in tackling IoT-specific obstacles such as the state explosion problem. The survey emphasizes the use of artificial intelligence (AI) in FV&V (Formal Verification and Validation) technologies and delineates future research paths to augment the dependability and safety of IoT systems [69].

As detailed in Table 3, the review of methodologies and applications of model checking in IoT systems encompasses

a range of IoT application fields, formal techniques, and primary topics. These studies illustrate the extensive use of model checking for verifying the correctness of Trickle-based dissemination protocols, ensuring the resiliency and sustainability of in-vehicle networks, validating the safety and security attributes of IoT systems, and assessing the performance of sensor networks. Moreover, the integration of model checking with deep learning and self-adaptation techniques has been explored to enhance security and flexibility in IoT systems.

The examined studies illustrate the substantial influence of model checking in verifying and guaranteeing the dependability, safety, and performance of IoT systems. Model checking has been successful in finding vulnerabilities, confirming protocol correctness, and assuring compliance with specified security properties by methodically examining the state space of complex systems. The incorporation of model checking with additional methodologies, such as deep learning and self-adaptation, amplifies its practicality and efficiency. With the rising complexity of IoT systems, the importance of model checking in assuring their reliability becomes more crucial. These studies emphasize the importance of continuous research and improvement in model checking approaches to tackle the changing difficulties in the IoT area.

C. THEOREM PROVERS

This subsection examines multiple research that have investigated the utilization of theorem provers in the context of IoT systems. It provides an overview of their techniques and findings. This review is organized in a chronological manner, starting with the earliest and progressing to the most recent, emphasizing significant contributions and achievements.

TABLE 4. Overview of recent research on theorem provers in IoT systems.

Study	Year	IoT Application Field	Formal Technique Used	Primary Topic	Main Contribution	Limitations
[70]	2022	Energy Systems	Interactive Theorem Proving	Verification	Validated energy distribution algorithms.	Complexity in large-scale energy systems.
[71]	2023	Formal Logic	Coq Theorem Prover	Formalization	Formalized propositional logic with meta-theorems.	Resource-intensive formalization process.
[72]	2023	Robotics	Coq Theorem Prover	Verification	Verified coordinate transformations and rotational matrices.	High computational requirements.
[73]	2023	Smart Contracts	Agda Theorem Prover	Verification	Created formal model for Solidity-style smart contracts.	Complexity in smart contract execution.
[74]	2023	Industrial IoT	Logic of Events	Security	Demonstrated security features of multi-factor authentication protocols.	Scalability and adaptability issues.
[75]	2024	IoT Ontologies	Logical-Algebraic Framework	Ontology Expansion	Enhanced interpretative capacities in qualitative spatial reasoning.	Complexity in model generalization and scalability.

An examination of future energy systems with interactive theorem proving, published in 2022. Ahmed et al. utilized interactive theorem proving to formally analyze prospective energy systems in this study [70]. The authors employed the Coq proof aid to validate the accuracy of energy distribution algorithms, guaranteeing the dependability and effectiveness of energy management in IoT-based smart grids. A rigorous formalization of propositional logic in Coq is presented in a study that specifically concentrated on the process of formalizing propositional logic using the Coq theorem prover. They have contributed to the field by creating four Hilbert-style axiom systems and a natural deduction system. Additionally, they have demonstrated equivalences and proven important meta-theorems, such as the Completeness and Soundness Theorems. This fundamental research in formal logic has practical implications in the verification of IoT systems, namely in guaranteeing the precision and dependability of logical operations within these networks [71].

In their study [72], the authors investigated the process of formally verifying robotic kinematics by utilizing the Coq proof aid. The work concentrates on verifying coordinate transformations and rotational matrices, which are crucial for accurately controlling robots in IoT environments. This study highlights the need of utilizing formal approaches to improve the safety and predictability of robotic operations in interconnected systems.

A formal representation of Solidity-style smart contracts in the theorem prover Agda, developed in 2023: the researchers employed the Agda theorem prover to create a model of Solidity-style smart contracts. Their objective was to establish a formalized and verified process for ensuring the accuracy of smart contract execution within IoT ecosystems [73]. This approach attempted to tackle the security and operational difficulties that are inherent in decentralized applications. Other study utilized the Logic of Events formal technique to demonstrate the security features of multi-factor authentication protocols in the context of industrial IoT.

The authors included novel event categories and axioms, which facilitated the rigorous examination and validation of protocol behaviors in the face of security risks [74].

In their 2024 study, the authors tackled the task of expanding formal ontologies in IoT systems. They showcased enhanced interpretative capacities and reasoning accuracy in qualitative spatial reasoning (QSR) theories by utilizing a logical-algebraic framework. This methodology facilitates the strong expansion of ontologies, which is crucial for the dynamic and scalable character of Internet of Things (IoT) applications [75]. In the same year, a new study introduced an algebraic recognition approach for identifying objects within the Internet of Things (IoT) ecosystem in the Aral area. The study devised accurate recognition algorithms utilizing linear closure operators, demonstrating their efficacy in rapidly identifying incoming information flows in IoT networks. As illustrated in Table 4, recent research on theorem provers in IoT systems highlights the diverse application fields and formal techniques used to address various primary topics such as verification, formalization, and security. The studies reviewed showcase significant contributions, including validating energy distribution algorithms, formalizing propositional logic with meta-theorems, verifying coordinate transformations and rotational matrices in robotics, creating formal models for Solidity-style smart contracts, demonstrating security features of multi-factor authentication protocols, and enhancing interpretative capacities in qualitative spatial reasoning for IoT ontologies.

The reviewed studies highlight the significant role of theorem provers in enhancing the reliability, security, and correctness of IoT systems. Through the formalization and verification of complex algorithms, protocols, and ontologies, theorem provers like Coq and Agda provide rigorous mathematical guarantees that are crucial for the dependable operation of interconnected devices. The integration of logical frameworks and algebraic methods further extends the applicability of theorem proving to diverse IoT scenarios,

TABLE 5. Summary of key studies on hybrid approaches in formal methods for IoT systems.

Study	Year	IoT Application Field	Formal Technique Used	Primary Topic	Main Contribution	Limitations
[76]	2020	Cloud Service Composition	Model Checking, Theorem Proving	Verification	Ensured service reliability and performance.	Complexity in large-scale systems.
[77]	2022	IoT Security	Trust Computing, Machine Learning	Security	Enhanced IoT security and data privacy.	Dependency on social metrics.
[78]	2020	Hybrid Systems	Component-Based Methods	Verification	Managed complexity of hybrid systems.	Scalability issues.
[79]	2023	Automotive Systems	Model Checking, Theorem Proving, Abstract Interpretation	Validation	Secured automotive systems.	Complexity in integration.
[80]	2021	Hybrid IoT OS	Formal Specification, Runtime Verification	Verification	Ensured system stability and reliability.	Resource-intensive.
[81]	2022	Hybrid Systems	Proof, Refinement with Event-B	Verification	Verified architectural patterns.	Flexibility constraints.
[82]	2021	Digital Circuits	Model Checking, Interactive Theorem Proving	Verification	Ensured circuit functionality.	High computational cost.
[83]	2021	Self-Driving Vehicles	MCMAS, PRISM Model Checkers	Decision-Making	Validated autonomous driving scenarios.	Scalability and adaptability.
[84]	2020	Avionics Systems	Model Checking, Multi-Agent Systems	Test Case Generation	Enhanced avionics safety and performance.	High computational requirements.
[41]	2022	Hybrid IoT Applications	Event-B, Generalization, Instantiation	Verification	Ensured correctness of hybrid IoT applications.	Complexity in model generalization.

from energy management and robotics to smart contracts and authentication protocols. These studies underscore the importance of continuing research and development in theorem proving techniques to address the evolving challenges in IoT, ensuring robust and secure systems in an increasingly connected world.

D. HYBRID APPROACHES

Hybrid approaches in formal methods for IoT systems integrate multiple techniques to leverage their individual strengths and mitigate their weaknesses. The selected studies illustrate the innovative ways in which researchers combine different formal methods to enhance verification, validation, and analysis processes in IoT systems.

In the study by Souri et al., a hybrid formal verification approach for QoS-aware multi-cloud service composition is proposed [76]. This approach combines model checking and theorem proving to ensure service reliability and performance. By integrating these methods, the authors were able to verify both functional and non-functional properties of cloud services, addressing issues such as resource allocation and service level agreements (SLAs). Liebrez et al. presented a service-oriented approach for decomposing and verifying hybrid system models. By using component-based methods, the study managed the complexity of hybrid systems in IoT. This approach enabled the modular verification of individual components, which could then be integrated into a complete system, ensuring that each part adhered to specified properties before being combined [77]. Elkholy et al. focused on model checking intelligent avionics systems using multi-agent systems for test case generation. This study blended formal methods with intelligent systems to enhance avionics safety and performance, ensuring that complex avionics software could be verified against stringent safety standards [78].

Guan et al. introduced a formal verification method for a hybrid IoT operating system model, combining formal specification languages with runtime verification. This method ensured system stability and reliability by allowing continuous verification of the system as it operates, thus catching potential errors and security issues in real-time [79]. Minhas et al. proposed HVoC, a hybrid model checking - interactive theorem proving approach for functional verification of digital circuits. This approach demonstrated the synergy between these methods in ensuring circuit functionality, particularly in verifying both the logical correctness and performance aspects of digital circuits used in IoT devices [80]. Al-Nuaimi et al. developed a hybrid verification technique for decision-making in self-driving vehicles. By combining MCMAS and PRISM model checkers, the study validated autonomous driving scenarios, ensuring that the vehicles could make safe and reliable decisions in various traffic conditions [81].

In 2022, Dupont et al. employed proof and refinement with Event-B to verify architectural patterns of hybrid systems. This study showcased the robustness of integrating multiple formal methods, allowing for the verification of complex system architectures while maintaining flexibility in design and implementation [82]. Similarly, Ali-Eldin et al. developed a hybrid trust computing approach using social similarity and machine learning. This method highlights the effectiveness of combining computational techniques with social metrics to enhance IoT security and data privacy. The study demonstrated how machine learning algorithms could be used to predict trust levels among IoT devices based on social similarities, thereby improving the overall security framework [83]. Lastly, Mammar et al. explored modeling and proving hybrid programs with Event-B, using generalization and instantiation techniques. This approach ensured the correctness of hybrid IoT applications by allowing for the

TABLE 6. Comprehensive comparison of focus areas in existing surveys vs. our survey.

Survey	Year	Focus Areas	Use of Formal Methods	Depth of Formal Methods Analysis	Limitations Addressed
[51]	2019	General IoT Applications	Mentioned Briefly	Low	Focus on lightweight cryptography, lacks detailed formal method discussion
[52]	2019	Real-Time Task Scheduling	Moderate	Moderate	Tool complexity in handling real-time IoT tasks
[53]	2020	Control Modules in CPS	Moderate	Low	Challenges in converting UML diagrams into verifiable models
[54]	2020	IoT Protocols	Limited	Moderate	General protocol verification, limited formal method exploration
[55]	2021	MQTT Protocol	Moderate	Moderate	Resource-intensive transformation process
[56]	2022	Smart Grids	High	High	Complexity in modeling energy distribution systems
[57]	2021	Safety-Critical IoT Systems	High	High	High computational requirements for timing analysis
[58]	2022	IoT Architecture	High	High	Resource-intensive design verification
[59]	2023	IoT Operating Systems	Extensive	High	Focused on scheduler verification, complex operational scenarios
[60]	2023	Constrained Devices	Moderate	Moderate	Limited exploration of formal verification beyond temporal logic
Our Survey	2024	IoT System Security and Reliability	Extensive	High	Detailed comparative analysis of formal methods, addressing gaps

verification of generalized models that could be instantiated for specific use cases, thus maintaining both flexibility and rigor in verification [84].

Krichen surveyed formal methods and validation techniques in the automotive industry, emphasizing the integration of model checking, theorem proving, and abstract interpretation. This comprehensive survey highlighted the importance of using multiple verification techniques to cover different aspects of automotive systems, such as safety and security, which are critical in IoT-enabled vehicles [41]. As shown in Table 5, hybrid approaches in formal methods for IoT systems encompass a range of application fields and integrate multiple formal techniques to enhance verification, validation, and security. The studies reviewed illustrate significant contributions, such as ensuring service reliability and performance in cloud service composition, enhancing IoT security and data privacy through trust computing and machine learning, managing the complexity of hybrid systems using component-based methods, securing automotive systems via a combination of model checking, theorem proving, and abstract interpretation, and ensuring system stability and reliability in hybrid IoT operating systems through formal specification and runtime verification.

To summary, these studies collectively demonstrate the effectiveness of hybrid approaches in addressing the complex verification and validation challenges in IoT systems. By integrating various formal methods, researchers can exploit the complementary strengths of these techniques, resulting in more robust, reliable, and secure IoT applications.

E. COMPARISON AND DISCUSSION

Several existing surveys on secure and reliable IoT systems have provided comprehensive insights into the challenges

and solutions from various perspectives. However, most of these studies tend to focus on general security mechanisms, cryptographic techniques, and network-level protections without addressing the deeper, model-driven formal methods for system verification and analysis. In contrast, our review emphasizes the necessity of applying formal methods to ensure that IoT systems meet stringent security and reliability requirements, especially in mission-critical applications such as healthcare, industrial IoT, and smart cities.

The 2019 survey [51] covers general IoT security with a focus on lightweight cryptographic techniques. While cryptography is vital for securing IoT communications, it does not provide the formal guarantees required to ensure the correctness of an IoT system under all operational conditions. Formal methods, which offer mathematically sound verification techniques, are only mentioned briefly in this survey. The depth of formal methods analysis is low, and the work does not thoroughly explore the broader applicability of formal verification.

In the same year, [52] analyzed real-time task scheduling in IoT, offering moderate coverage of formal methods. The complexity of verifying real-time tasks, where timing constraints are critical, presents a significant challenge. The moderate depth of formal analysis reflects this, as formal verification techniques such as model checking and theorem proving are powerful but can be resource-intensive when applied to real-time systems. Similarly, [54] examined the verification of IoT protocols, focusing on a limited subset of formal methods. The study explored how formal methods could be used to verify protocol behavior, but the depth of analysis was moderate, with limited exploration of scalability and real-time concerns.

Surveys such as [57] and [58], conducted in 2021 and 2022 respectively, delve deeper into formal methods by

applying them to safety-critical IoT systems and IoT architecture. These surveys highlight the importance of using formal verification techniques such as model checking and theorem proving to ensure the correctness and timing behavior of safety-critical systems, where errors can lead to catastrophic outcomes. Despite the high depth of analysis, these works also face limitations in computational overhead and resource-intensive verification processes.

The study by [59], conducted in 2023, focuses extensively on formal methods, particularly in verifying the Contiki scheduler used in IoT operating systems. By extracting and verifying scheduler operations using formal specifications, this work exemplifies how formal methods can ensure system stability and correct task execution in complex operational scenarios. Although this survey provides extensive coverage of formal methods, it still encounters challenges with verifying the complexity of scheduler operations.

While [60] in 2023 covers constrained IoT devices and uses temporal logic for multi-factor authentication verification, it offers only moderate depth in its formal analysis. This reflects a common limitation in existing surveys, where formal methods are applied to a narrow subset of security challenges but are not fully explored as a comprehensive solution for IoT reliability and scalability. As shown in Table 6, existing surveys on secure and reliable IoT systems have varying degrees of focus on formal methods, with some only briefly mentioning their applicability while others delve deeper into specific use cases like safety-critical IoT systems or protocol verification.

In contrast to the existing surveys, our review provides an extensive and detailed analysis of formal methods, particularly focusing on their application to IoT security and reliability. Our survey dives deeply into various formal techniques, such as model checking, theorem proving, and formal specification languages, to address critical IoT challenges such as scalability, real-time performance, and system complexity. By doing so, we address the gaps found in previous surveys, offering a comprehensive view of how formal methods can ensure the robustness and security of IoT systems.

Our survey contributes significantly by not only identifying the challenges of existing approaches but also providing solutions through formal verification methods. In particular, our work highlights recent advancements and emerging trends, such as the integration of hybrid formal methods with runtime verification techniques and machine learning, which are not extensively covered in earlier works.

IV. KNOWLEDGE GAPS

Although the literature on the use of formal methods in IoT systems has shown great progress, there are still some areas where our understanding is lacking. To improve the dependability, safety, and efficiency of IoT systems, it is crucial to address these deficiencies. The following critical areas have been highlighted where additional research and development are imperative.

A. SCALABILITY ISSUES

A key obstacle consistently found in several research is the ability of formal approaches to handle the size and complexity of IoT systems. As the Internet of Things (IoT) networks expand, the number of possible states that need to be examined using formal verification methods like model checking increases rapidly. This sometimes results in a state space that becomes too large to handle computationally, making it infeasible to perform the necessary calculations. Furthermore, scalable formal methods and verification tools are required to effectively handle large-scale IoT systems while maintaining accuracy and performance. Subsequent investigations should prioritize the development of algorithms and procedures capable of managing the growing intricacy and magnitude of IoT networks, as Figure 8 shows the scalability issues and state space explosion.

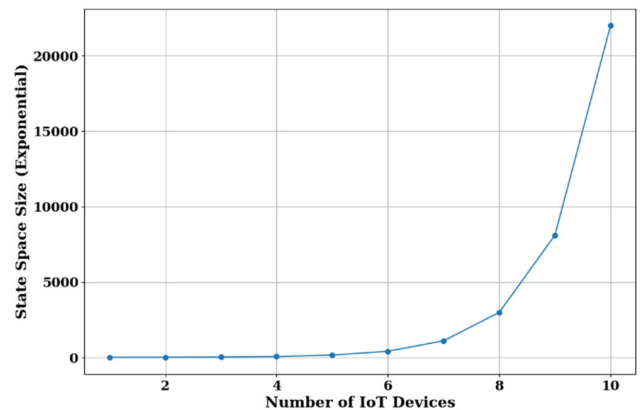


FIGURE 8. Exponential growth of state space size with increasing number of IoT devices.

The diagram depicts the scalability problem referred to as state space explosion in the domain of formal verification of Internet of Things (IoT) systems. The x-axis denotes the quantity of IoT devices, while the y-axis illustrates the magnitude of the state space on an exponential scale. Every point on the graph represents the extent of the state space that corresponds to a specific number of IoT devices. Furthermore, the graph illustrates that the size of the state space increases exponentially as the number of IoT devices increases. At first, when there are fewer devices, the size of the state space is modest and easy to handle. Nevertheless, as the quantity of devices grows, the dimension of the state space expands rapidly, reaching a point where it becomes quite extensive and challenging to manage. The rapid and exponential expansion poses a considerable obstacle for formal approaches, such as model checking, as they heavily depend on the exploration of the state space to validate system features. The exponential growth of the state space size necessitates a corresponding rise in computational resources, such as memory and processing power, for verification purposes. Consequently, it becomes impossible to verify systems that involve a substantial number of devices.

The consequences for IoT systems are especially serious, as these systems frequently consist of a substantial number of networked devices, which makes them vulnerable to state space explosion. This emphasizes the need for verification methodologies that can be scaled up and effectively handle and reduce the impact of state space explosion. The illustration emphasizes the significance of creating adaptable formal techniques capable of managing the intricacy of extensive IoT networks. Abstraction, compositional verification, and symbolic methods are essential techniques for effectively handling the problem of state space explosion.

The diagram also highlights the necessity of utilizing sophisticated verification techniques that integrate optimizations and heuristics to decrease the size of the state space while maintaining correctness. These techniques can assist in maintaining the feasibility of formal verification as IoT systems get more intricate. Future research should prioritize improving the scalability of formal methods, investigating hybrid approaches that integrate various verification techniques, and constructing frameworks capable of effectively managing extensive state spaces. Exploring domain-specific improvements designed for IoT applications might enhance the efficiency and feasibility of verification processes. To assure the reliability, security, and performance of IoT systems, researchers and practitioners should address the scaling concerns depicted in the figure, thus allowing formal approaches to offer credible guarantees.

In summary, scalability remains a significant challenge for IoT systems as the number of connected devices and the volume of data continues to grow exponentially. Formal methods, while useful, struggle to manage the large and complex state spaces that emerge in large-scale IoT networks. This limitation often results in bottlenecks that impact the overall efficiency and reliability of these systems, highlighting the need for scalable formal verification techniques.

B. CONSTRAINTS ON REAL-TIME PERFORMANCE

Several IoT applications, particularly those in safety-critical fields such as healthcare and industrial automation, adhere to stringent real-time limitations. Although there has been research on real-time task scheduling, there is a shortage of complete formal techniques that can ensure real-time performance across a wide range of IoT applications. There is a need for more study to provide formal techniques that can successfully verify and enforce real-time limitations in IoT systems. This involves ensuring that essential tasks are completed within their specified timeframes and that the system maintains a consistent and dependable performance in all operational circumstances. Figure 9 shows the real-time constraints and task scheduling.

The diagram depicts the time limitations that exist when scheduling tasks in IoT systems. The x-axis corresponds to the temporal dimension, whereas the y-axis reflects distinct activities or assignments. Each colored bar represents the duration of a task, and certain deadlines are indicated by vertical red dashed lines. Moreover, the diagram depicts the

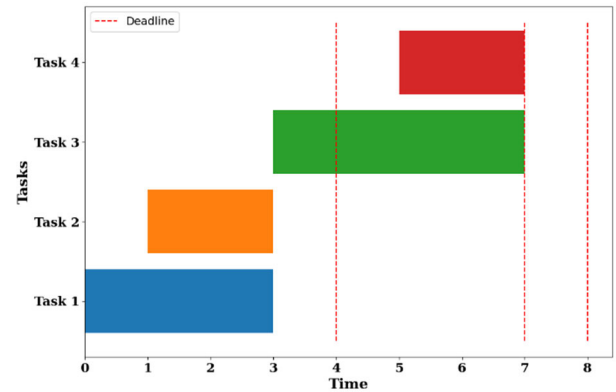


FIGURE 9. Real-time constraints and task scheduling in IoT systems.

chronological arrangement of four tasks, labeled as Task 1 to Task 4. Every work is assigned a distinct start and end time, which is visually depicted by the duration of the colored bars. Task 1 commences at time 0 and concludes at time 3. Task 2 commences at time 1 and concludes at time 3. Task 3 commences at time 3 and concludes at time 6. Task 4 commences at time 5 and concludes at time 7. The deadlines for Task 3 and Task 4 are indicated by vertical red dashed lines at time 4 and time 7, respectively. Task 3 exceeds its deadline, suggesting a failure to fulfill the deadline and a potential violation of real-time constraints, but Task 4 completes precisely on time, successfully adhering to the real-time restriction.

Ensuring real-time limitations are met is crucial in IoT systems, particularly for tasks that are time-sensitive or have implications for safety and mission-critical operations. The diagram illustrates the significance of effective task scheduling in order to guarantee the timely completion of all jobs. The successful coordination of Task 1 and Task 2 demonstrates that by implementing effective scheduling, tasks may be accomplished within their designated time periods. Nevertheless, the failure to fulfill the deadline for Task 3 underscores the challenges of assuring timely completion of all tasks, particularly when there is task overlap or dependency on the completion of preceding tasks.

The challenge of maintaining real-time performance in IoT systems is critical, particularly in sectors like healthcare and industrial automation. Delays in data processing and network latency often compromise the ability of IoT systems to meet strict timing constraints. Currently, formal methods do not fully address these real-time performance issues, which limits their applicability in time-sensitive IoT applications where precision is essential.

C. INCORPORATION OF VARIED FORMAL METHODS

Various formal methods, such as model checking, theorem proving, and formal specification, provide distinct advantages and constraints. Nevertheless, there is a scarcity of study on the integration of these various strategies to exploit their complementing benefits. Integrating different components could offer stronger and more thorough verification solutions

for Internet of Things (IoT) systems. It is crucial to explore strategies for smoothly combining several formal methodologies. The research should prioritize the integration of several methods to effectively tackle all aspects of IoT system needs, encompassing modeling, verification, and analysis. Figure 10 presents the integration levels of various formal methods in IoT systems.

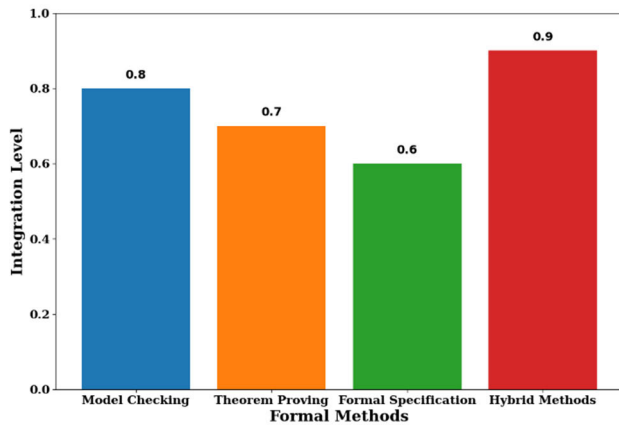


FIGURE 10. Integration levels of various formal methods in IoT systems.

The diagram depicts the levels of integration of different formal approaches inside the framework of IoT systems. The x-axis indicates several categories of formal methods, while the y-axis represents the degree of integration on a scale ranging from 0 to 1. Every bar in the chart represents a distinct formal technique and indicates its degree of integration within IoT systems. The chart displays four distinct formal methods: Model Checking, Theorem Proving, Formal Specification, and Hybrid Methods. The integration levels of Model Checking, Theorem Proving, Formal Specification, and Hybrid Methods are 0.8, 0.7, 0.6, and 0.9, respectively. These findings suggest that Hybrid Methods are the most widely incorporated into IoT systems, followed by Model Checking, Theorem Proving, and Formal Specification.

The extensive integration level of Hybrid Methods implies that the combination of several formal methods can offer a more thorough and efficient way to validating and analyzing IoT systems. Hybrid methods utilize the advantages of numerous techniques, providing a strong framework for tackling the intricacies and difficulties of IoT systems. Model Checking, with an integration level of 0.8, is extensively utilized because of its automated verification capabilities. These capabilities enable the systematic investigation of the state space of IoT systems to ensure that they satisfy specified attributes. Theorem Proving, with an integration level of 0.7, is well regarded for its capacity to deliver meticulous mathematical proofs, guaranteeing the accuracy of system behavior. Formal Specification, at an integration level of 0.6, is crucial for accurately delineating system requirements and behavior, enabling early identification of design flaws.

While various formal methods such as model checking and theorem proving have been applied to IoT systems,

their integration remains insufficient. The lack of comprehensive frameworks that combine these different techniques limits their ability to effectively address the wide range of challenges faced by IoT systems. This gap hinders the development of holistic solutions that can manage both functional and non-functional requirements across diverse IoT architectures.

D. ADAPTABILITY AND EVOLUTION

IoT systems are intrinsically characterized by their dynamic nature, as devices and configurations undergo regular changes over time. Contemporary formal techniques typically rely on static system models, neglecting the dynamic character of IoT contexts. There is a requirement for flexible formal procedures that can adapt to modifications in system architecture and behavior. It is necessary to develop adaptive formal methods that can dynamically update verification models in response to changes in the IoT system. This will guarantee ongoing accuracy and dependability as the system progresses. Figure 11 depicts the flexibility and progression of IoT systems, highlighting the ever-changing nature of these systems where devices and configurations experience frequent modifications over time. The x-axis denotes the temporal dimension, while the y-axis signifies a range of evolutionary occurrences. Distinct colors are employed to signify several categories of events: blue denotes the addition of a device, green represents updates to a device, and orange indicates network reconfigurations.

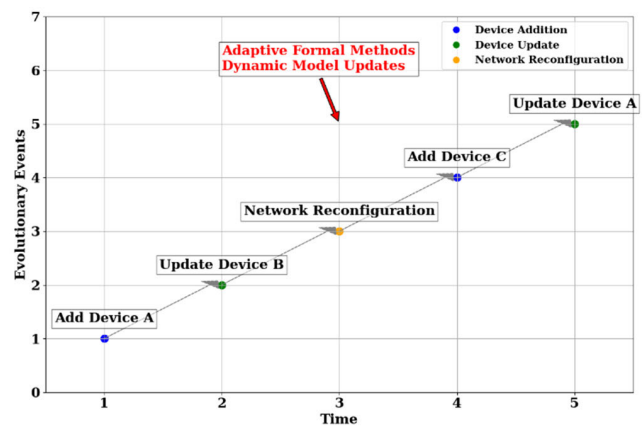


FIGURE 11. Adaptive formal methods and evolutionary events in IoT systems.

The chronology depicts five crucial occurrences: the inclusion of Device A, the enhancement of Device B, a reconfiguration of the network, the addition of Device C, and the update of Device A. Every event is indicated by a colored dot, accompanied with accompanying annotations that offer clarification on the nature of each occurrence. The arrows connecting the events illustrate the direction and chronological order of evolution. The dashed lines symbolize the advancement and interrelation of various events. The arrows

emphasize the ongoing development of IoT systems, characterized by regular upgrades and reconfigurations.

The graphic prominently emphasizes the importance of adaptable formal approaches by a big annotation at the top. The annotation “Adaptive Formal Methods Dynamic Model Updates” underscores the necessity for formal methods that possess the ability to dynamically adjust to the ever-changing IoT system. The significance of these adaptive strategies in preserving accuracy and reliability as the system progresses is emphasized by the prominent arrow and red text.

The ever-changing characteristics of IoT systems require the creation of flexible formal procedures that can adapt to real-time modifications. Static formal models are inadequate for IoT contexts due to the continuous evolution of system configurations. The graphic aptly demonstrates the dynamic character of the subject matter by visually presenting the sequence of significant events over time and their interconnections. The necessity for adaptability in formal approaches is emphasized by the frequent and frequently unforeseeable alterations in IoT contexts. This includes the incorporation of novel devices, modifications to current devices, and alterations in network setups. Adaptive formal techniques need to be developed in a way that allows for smooth integration of these changes into the verification process, while guaranteeing that the system’s reliability and security are preserved.

Adaptive formal methods play a significant role in maintaining the accuracy and reliability of verification procedures by reflecting the current state of the system. These methods ensure the continuing accuracy and reliability of IoT systems by updating verification models in response to changes. To guarantee the accuracy, reliability, and security of IoT systems as they evolve, it is crucial to develop adaptive formal methods that can dynamically update verification models in response to system changes. This technique overcomes the constraints of static formal methods and effectively tackles the distinct issues presented by the dynamic nature of IoT contexts.

IoT systems must be capable of evolving and adapting to dynamic environments where devices are frequently added, updated, or removed. However, current formal methods are not fully equipped to handle this need for continuous adaptation without requiring complete system revalidation. This limits the flexibility of IoT systems, particularly in fast-changing environments where adaptability is crucial for maintaining system functionality and stability.

E. SECURITY AND PRIVACY

While several studies prioritize the verification of functional correctness, there is relatively less attention given to the elements of security and privacy. Due to the importance of data sensitivity and the potential consequences of security breaches in IoT systems, there is a pressing requirement for formal approaches that expressly focus on addressing these concerns. It is crucial to enhance formal approaches in order to thoroughly test security and privacy features in IoT systems. This entails the development of methodologies to

safeguard against attacks and illegal entry, as well as guaranteeing the resilience and dependability of security methods. Figure 12 depicts the significance of security and privacy in IoT systems, emphasizing its crucial purpose in ensuring the overall integrity and performance of these systems. The x-axis indicates two fundamental dimensions: Security and Privacy, while the y-axis displays the proportion.

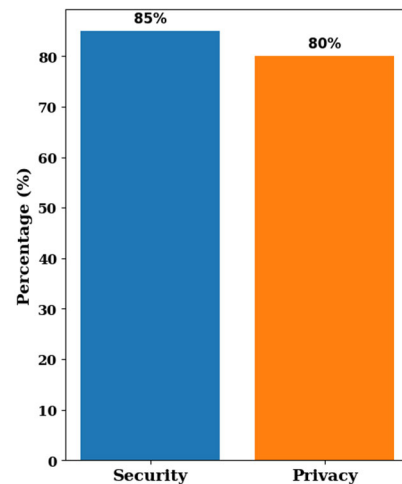


FIGURE 12. Security and privacy in IoT systems.

The bar chart illustrates that security is valued at 85% and privacy at 80%, showing their significant significance. The graphic illustrates the essentiality of both security and privacy in the development and execution of IoT systems. The 85% security grade highlights the importance of safeguarding IoT systems against a range of dangers, such as illegal access, data breaches, and cyberattacks. Security measures are crucial for guaranteeing the secrecy, consistency, and accessibility of data within IoT networks.

Furthermore, the 80% grade on privacy underscores the significance of protecting personal and sensitive data that is gathered and transmitted by IoT devices. The importance of privacy is of utmost significance, particularly in light of the growing volume of data produced by Internet of Things (IoT) devices across a range of applications, including smart homes, healthcare, and industrial automation. Ensuring privacy necessitates the implementation of safeguards to prevent illegal access and exploitation of data, while also complying with legislative mandates and meeting user expectations. The similar levels of security and privacy percentages indicate that both factors are given equal priority in IoT systems. The maintenance of this equilibrium is of utmost importance as robust security measures have the potential to augment privacy, while efficient privacy practices can fortify overall security.

F. USER-FRIENDLY TOOLS

The intricacy of formal approaches frequently constrains their implementation in pragmatic IoT development. There is a

requirement for tools that are easy for users to understand and use, which can simplify the complicated mathematical aspects and offer straightforward interfaces for developers and engineers. It is essential to develop formal verification tools that are more accessible and user-friendly in order to encourage more use among practitioners. These technologies should be seamlessly included into the IoT development process and offer tangible advantages without necessitating extensive knowledge in formal methodologies. Focusing on three main criteria—Ease of Use, Intuitive Interface, and Documentation—Figure 13 shows the user evaluations for tools applied in formal techniques. The x-axis indicates these features; the y-axis displays the ratings ranging from 5.

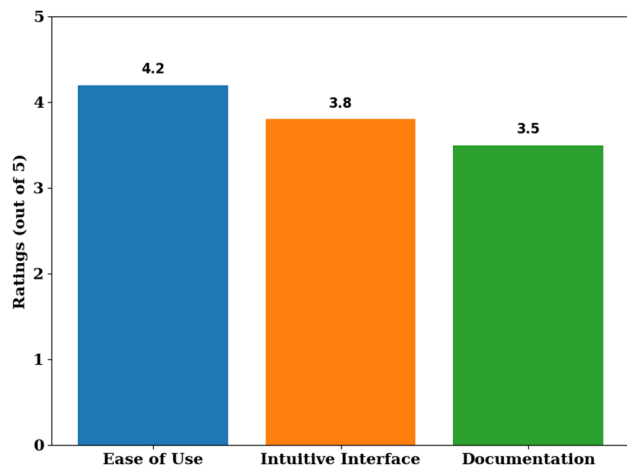


FIGURE 13. User-friendly tools for formal methods.

In fact, users believe these tools to be of 4.2 for Ease of Use, 3.8 for Intuitive Interface, and 3.5 for Documentation, therefore reflecting their perceived quality and usability. With a rating of 4.2, the chart shows that Ease of Use is greatest implying that consumers find formal methods tools rather simple to use. This high grade emphasizes the need of creating tools that are easily available and user-friendly so that practitioners may efficiently implement formal approaches free from too complex or steep learning curves. Although the tools are usually regarded as user-friendly, the 3.8 Intuitive Interface rating shows that there is still space for development in simplifying interfaces. Since it improves user experience, lowers the possibility of mistakes, and improves tool efficiency in formal verification and analysis, an easy interface is absolutely vital.

Though rather lower, the 3.5 Documentation rating emphasizes the requirement of thorough and unambiguous documentation. Users need good documentation if they are to grasp the tools' features, capacities, and limits. It supports learning, troubleshooting, and efficient application of the functionality of the products. The somewhat low-grade points to an area that needs work since it implies that users could be having trouble locating sufficient tools and help.

G. FORMAL METHODS FOR EMERGING IoT TECHNOLOGIES

With the emergence of new IoT technologies like edge computing, 5G, and AI-driven IoT, there is a lack of formal methodologies being applied to these technologies. The current research predominantly concentrates on conventional IoT systems, neglecting the exploration of future technologies. It is crucial to extend the use of formal approaches to new IoT technologies. The research should prioritize the development and adaptation of formal methodologies to test and analyze emerging IoT paradigms, with a specific emphasis on meeting performance, security, and reliability standards. Figure 14 depicts the hierarchical levels at which formal approaches are applied in three developing IoT technologies: AI-driven IoT, 5G, and Edge Computing. The x-axis depicts the level of implementation, ranging from 0 to 5, while the y-axis displays the three technologies. The bars represent the level of implementation of formal techniques in each technology, with AI-driven IoT ranked at 4.8, 5G at 4.2, and Edge Computing at 3.5.

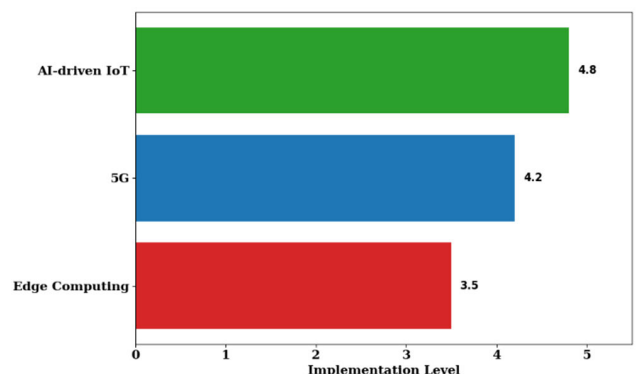


FIGURE 14. Formal methods for emerging IoT technologies.

According to the Figure 13, AI-driven IoT has the highest level of implementation of formal procedures, scoring 4.8. Formal approaches are widely used to guarantee the reliability, security, and efficiency of AI-driven IoT systems at a high level. Due to the intricate and crucial nature of AI algorithms in IoT applications, the meticulous use of formal methods aids in proving the accuracy and resilience of these systems, hence increasing trust and reliability. The 5G technology, which has an implementation level of 4.2, also demonstrates substantial utilization of formal methods. Deploying 5G networks requires intricate protocols and instantaneous data transmission, which makes it necessary to employ formal methods for validating and verifying these systems. Formal approaches in the context of 5G technology are employed to guarantee that the demanding criteria of high speed and low latency are achieved, all the while upholding security measures and averting possible weaknesses. Edge Computing, with a rating of 3.5, exhibits a somewhat lower although still significant degree of formal techniques implementation. Edge Computing entails the processing of data in close

proximity to its origin, as opposed to depending on centralized cloud servers. This approach has distinct issues in terms of limited resources and the need for real-time processing. Applying formal approaches in Edge Computing offers a systematic approach to ensuring the accuracy and efficiency of distributed systems that operate at the edge of the network.

While formal methods provide valuable tools for enhancing the security, reliability, and scalability of IoT systems, their effectiveness varies across different identified knowledge gaps. As depicted in Figure 15, formal methods show strong coverage in areas like security and privacy but are less capable of addressing challenges related to scalability, adaptability, and the evolution of IoT systems. This highlights the need for further development and integration of formal methods to better meet the demands of rapidly evolving IoT technologies.

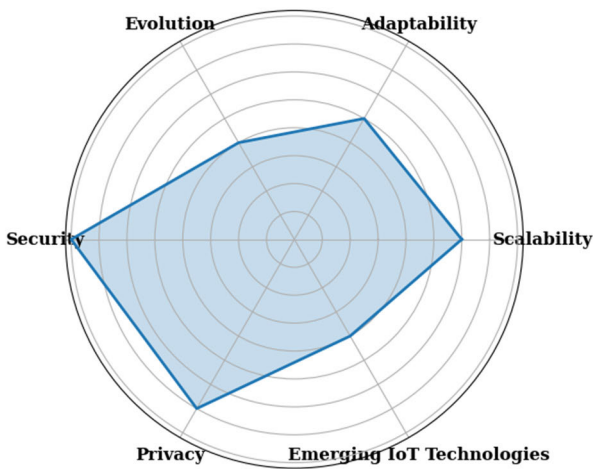


FIGURE 15. Coverage of formal methods in addressing IoT knowledge gaps.

This radar chart illustrates how well formal methods currently address each major knowledge gap, emphasizing the areas where formal methods are effective (e.g., security, privacy) and where improvements are needed (e.g., scalability, adaptability).

V. CHALLENGES AND FUTURE CONSIDERATIONS

The application of formal methods to IoT systems has demonstrated significant potential for enhancing system security and reliability. However, several challenges remain that hinder their broader adoption and practical implementation. Addressing these challenges is crucial for advancing the field and ensuring that IoT systems are robust, secure, and scalable.

One of the primary challenges in applying formal methods to IoT systems is scalability. As IoT networks grow, the state space that needs to be explored during verification increases exponentially, resulting in what is known as the state space explosion problem. Current formal methods are limited by computational resources, making it difficult to handle the verification of large-scale IoT networks. While abstraction techniques have been proposed to mitigate this issue, further

research is needed to develop more scalable algorithms. Optimizing formal verification methods to handle the complexity of large and distributed IoT networks is essential to ensure their continued reliability as they expand. The current depth of formal methods applied to scalability challenges is still insufficient and is visualized in Figure 16, indicating the need for further development in this area.

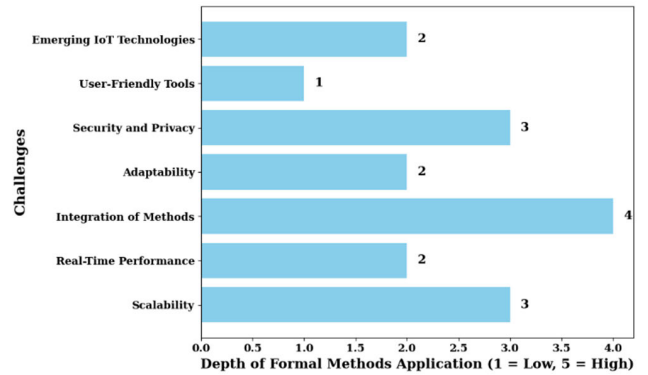


FIGURE 16. The current depth of formal methods application across various IoT challenges demonstrating the areas where further development is needed.

Another critical issue is ensuring real-time performance in IoT systems. Many IoT applications, particularly those in safety-critical domains like healthcare and industrial automation, require tasks to be executed within strict time constraints. However, existing formal methods are not fully equipped to verify real-time performance across a wide range of dynamic and resource-constrained environments. Although some progress has been made in real-time task scheduling, more comprehensive formal verification tools are needed to guarantee real-time performance while maintaining system efficiency. As shown in Figure 1, the depth of formal methods in addressing real-time constraints remains moderate, suggesting that additional research is required.

The integration of varied formal methods, such as model checking, theorem proving, and formal specification languages, presents another opportunity for advancing the field. Each method has distinct strengths, but they also have limitations when used in isolation. For example, while model checking provides automatic verification, it often struggles with scalability. In contrast, theorem proving can handle more complex systems but requires significant manual effort. There is a clear need for hybrid verification frameworks that combine the strengths of these methods to offer a more comprehensive and efficient approach to system verification. The importance of this integration is reflected in Figure 17, which highlights it as a priority for future research.

This radar chart illustrates the research priorities for formal methods in IoT systems, emphasizing areas such as scalability, real-time performance, security, and the integration of methods. The numerical values represent the relative importance of each research area on a scale of 1 to 5, where 5 indicates the highest priority.

TABLE 7. Overview of proposed future research directions.

Future Research Area	Research Focus	Expected Impact
Scalable Formal Methods	Development of techniques like abstraction and compositional verification to handle large-scale IoT systems.	Address scalability challenges in complex IoT networks.
Real-Time Formal Verification	Creation of comprehensive real-time verification tools for dynamic and resource-constrained IoT environments.	Ensure real-time performance for safety-critical IoT applications.
Hybrid Formal Methods	Integrating model checking, theorem proving, and other formal methods for a more comprehensive verification approach.	Improve overall system reliability by combining method strengths.
Adaptive Formal Methods	Flexible verification techniques that update in real-time with evolving IoT architectures.	Enhance verification for dynamic and evolving IoT environments.
Security and Privacy Frameworks	Formal methods that rigorously verify security protocols and privacy measures.	Strengthen IoT system resilience to security attacks and data breaches.
User-Friendly Formal Tools	Development of intuitive and accessible verification tools for IoT practitioners.	Broaden the adoption of formal methods in practical IoT applications.
Formal Methods for Emerging IoT Technologies	Extending formal verification techniques to new paradigms like edge computing, 5G, and AI-driven IoT systems.	Ensure future IoT technologies are reliable and secure.

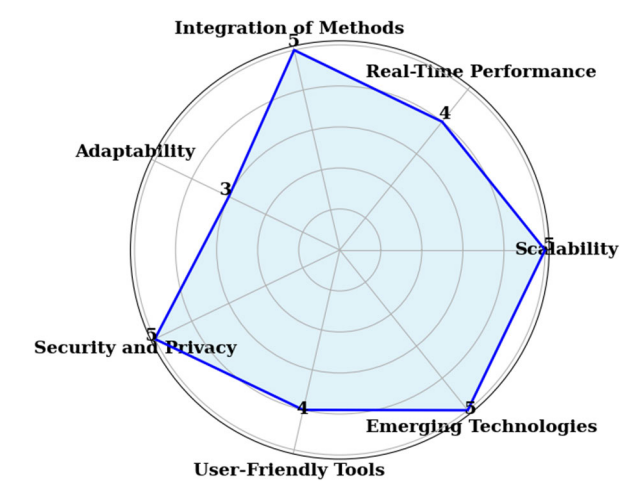


FIGURE 17. Future research priorities for formal methods in IoT.

Adaptability is also a significant concern for IoT systems, which are inherently dynamic. Devices and system architectures frequently change over time, making it challenging for existing formal methods, which often rely on static models, to keep pace with evolving systems. Developing adaptive formal methods that can dynamically update verification models as systems evolve is critical for maintaining the reliability of IoT systems in real-time environments. This area remains underexplored, as seen in Table 7, where adaptive formal methods are identified as a key research direction.

Security and privacy are paramount in IoT systems, particularly given the sensitive nature of the data they handle and the potential consequences of security breaches. While formal methods have traditionally focused on functional correctness, there is a growing need for methods that rigorously verify security protocols and privacy policies. Ensuring that IoT systems can withstand security attacks while maintaining data confidentiality and integrity is an essential requirement for future IoT networks. Formal methods that address these

concerns are still limited, as indicated by their moderate depth in Figure 16.

Despite the potential of formal methods, their complexity often limits their adoption in practical IoT development environments. Current formal verification tools require advanced knowledge of mathematical theories, making them inaccessible to non-expert developers and practitioners. Developing user-friendly formal tools that integrate seamlessly into IoT development workflows is crucial for broader adoption. Such tools should provide intuitive interfaces and extensive documentation without compromising the rigor of the underlying formal methods. As highlighted in Table 7, the development of user-friendly tools represents a significant opportunity for expanding the practical use of formal methods in IoT systems.

Lastly, the emergence of new IoT technologies, such as edge computing, 5G, and AI-driven IoT systems, presents both challenges and opportunities for formal methods. While much of the current research focuses on traditional IoT architectures, these new technologies introduce additional layers of complexity and uncertainty. Extending formal verification techniques to cover these emerging paradigms will ensure that next-generation IoT systems meet the same high standards of reliability, security, and scalability as traditional systems.

VI. CONCLUSION AND PERSPECTIVES

The exhaustive examination of formal approaches employed in IoT systems highlights their pivotal function in augmenting the dependability, security, and efficiency of networked devices. The utilization of formal specification, model verification, and theorem proving has played a crucial role in tackling the inherent difficulties of IoT systems. These approaches have shown substantial promise in confirming protocol consistency, ensuring real-time speed, and offering mathematical assurances for the security of IoT systems. However, the assessment emphasizes that there are still obstacles that need to be addressed. These challenges include

scalability limitations, real-time constraints, integration of multiple formal approaches, adaptability, security and privacy concerns, and the need for better user-friendly tools. Tackling these obstacles is crucial for the ongoing advancement and execution of dependable IoT systems.

In order to address the deficiencies that have been highlighted and improve the use of formal methods in IoT systems, it is recommended that future research concentrate on creating verification tools that can handle large-scale IoT networks and real-time restrictions in a scalable manner. The integration of various formal approaches can enhance their complementing capabilities, resulting in more resilient verification systems. Furthermore, the implementation of adaptive formal methods that can dynamically modify verification models in accordance with alterations in system configurations would guarantee uninterrupted dependability. By implementing thorough verification procedures and developing user-friendly tools, the adoption of the system among practitioners would be facilitated, leading to improved security and privacy. Moreover, by extending the use of formal methods to emerging technologies like edge computing, 5G, and AI-driven IoT, we can guarantee that these new approaches adhere to performance, security, and reliability criteria, thus facilitating the development of the next generation of IoT systems.

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